

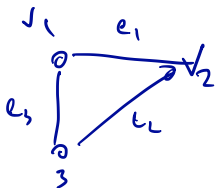
Bipartite Graphs and TUM

Thm: $G = (V, E)$

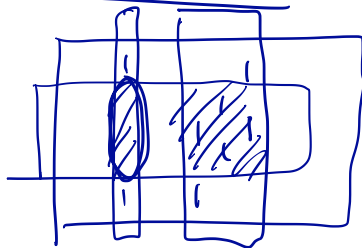
A $V \times E$ incidence matrix

$$A_{v,e} = \begin{cases} 1 & \text{if } e \in \delta(v) \\ 0 & \text{o.w.} \end{cases}$$

A is TU iff G is
bipartite.



	e_1	e_2	e_3
v_1	1	0	1
v_2	1	1	0
v_3	0	1	1



The proof of one direction.

(If G is bipartite $\Rightarrow A \in TU$).

proof is by induction on the size of
square submatrix. 1×1 immediate.

Let $\hat{A}$ be a $k \times k$ submatrix of A

- If any col of $\hat{A}$ is all zeros then $\det \hat{A} = 0$.
done.
- If any column of $\hat{A}$ has a single 1, then by exp'n by
minors along that column + ind. hypothesis, done.
- If all columns of $\hat{A}$ have 2 1's.

Since G bipartite, can partition rows into 2 groups, so that each column
of $\hat{A}$ has a 1 in 1st group and a 1 in 2nd group.

$$\Rightarrow \hat{A} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \det \hat{A} = 0 \text{ (square)}$$

$$\hat{A} = \begin{bmatrix} \begin{matrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & 1 & \\ & & & & 1 \end{matrix} \\ \hline \begin{matrix} & & & & 1 \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{matrix} \end{bmatrix}$$

Generalizing König — 1/2

Revisit the min vertex cover

Max Matching LP

$$\max: \sum x_e$$

$$\text{st: } \sum_{e \in \delta(v)} x_e \leq 1, \forall v$$

$$x \geq 0$$

Min Vertex Cover LP

$$\min: \sum y_v$$

$$\text{st: } y_{v_1} + y_{v_2} \geq 1 \quad \forall (v_1, v_2) \in E$$

$$y \geq 0$$

by above: TU. (b/c G is assumed bipartite)

by Hoffman - Kruskal Thm from previous lecture,

Min vertex cover LP is integral.

$\Rightarrow$ König's Thm follows from
TU + Duality.

Generalizing König — 2/2

For b, c an integral vector,

a b -matching is an assignment

$$x: E \rightarrow \mathbb{Z}_{\geq 0} \quad \text{where each}$$

vertex can be covered up to b_v times.

$$\sum_{e \in \delta(v)} x_e \leq b_v$$

A c -vertex cover is an assignment: $y: V \rightarrow \mathbb{Z}_{\geq 0}$

$$y_{v_1} + y_{v_2} \geq c_{(v_1, v_2)}$$

Thm: For any bipartite graph, b, c integral vec
the max c -weighted b -matching is equal to the min b -weighted c -vertex cover.

M square.

M TUM $\Leftrightarrow M, M^{-1}$ entries in $\{-1, 0, +1\}$.